

Course Information

Course NAME: **MGMT 59000 AI for Business Analytics – DY2**

Course CRN: 11420

Semester & Module Dates: Summer 2026, 06/29/2026 - 08/16/2026

Instructional Modality: Asynchronous Online

Course Credit Hours: 2

Instructor: Juan Ignacio Gonzalez-Espinosa, Ph.D.

E-mail: gonz1400@purdue.edu

Office: KRAN 533

Weekly Synchronous session: Every Tuesday 7:00-8:30 PM EST

Online Office Hours: Every Thursday 7:00-8:30 PM EST

Zoom link for synchronous activity and office hours:

<https://purdue-edu.zoom.us/j/92110217860?pwd=2LXy2WDQsUnEungONf17WyFP8nP2mH.1>

Meeting ID: 921 1021 7860

Passcode: 658291

Teaching Assistant: Himanshu Laddhad

E-mail: hladdhad@purdue.edu

Course Description

In this course, students explore and learn how artificial intelligence empowers the development of data-driven solutions supporting business decision-making. You will integrate AI tools, programming environments, and analytical techniques to design and deploy intelligent solutions for real business challenges. Emphasizing experimentation, teamwork, and creative problem-solving, the course equips students with the technical and foundational analytical skills needed to transform data into actionable insights and to thrive in AI-enhanced decision environments.

Learning outcomes:

Upon completing this course, students will be able to:

- *Apply diverse analytics platforms and programming languages to process, visualize, and interpret business data.*
- *Use artificial intelligence tools, including generative AI and low/no-code environments, to create data-driven business solutions.*

- *Collaborate to solve business challenges* with analytical rigor, creativity, and strategic insight.
- *Communicate analytical results through clear, professional reports and presentations* for academic and business audiences.
- Design an organized professional portfolio website showcasing projects, code, and posters on GitHub.

Prerequisites: MGMT 670 Business Analytics or an equivalent statistics course.

Technical Requirements

This course uses widely used analytics and AI-enabled tools to support hands-on, applied learning in an online environment. Core software includes:

- ✓ RStudio (personal desktop download) or Purdue's cloud-based Scholar-Gateway
- ✓ Python (Jupyter notebook, personal desktop download) or Purdue's cloud-based Scholar-Gateway
- ✓ Google Colab
- ✓ OpenAI, Copilot, Gemini, Claude, and Purdue GenAI Studio.

Computing (Required): A laptop or desktop computer with internet access and the capability to run Python and R programs. Ensure your computer meets the following desired requirements:
Processor (CPU): Intel i5 (8th gen or newer) / AMD Ryzen 5 (2000 series or newer). RAM: 8 GB.
Operating System: Windows 10/11, macOS 12+, or Linux (Ubuntu 20+).

Textbooks and readings (optional)

This course includes a variety of reference materials across its modules. Although these resources are not required, students are strongly encouraged to review them to deepen their understanding of each topic covered in the course.

1. **(ISL)** An Introduction to Statistical Learning. Second Edition. James, Witten, Hastie, and Tibshirani. <https://www.statlearning.com/>
2. **R for Everyone:** Advanced Analytics and Graphics (2nd Edition), by Lander, et al. ([ISBN-13:978-0134546926](#)). Available through the Purdue Library.
3. **Python for Data Analysis:** Data Wrangling with pandas, NumPy, and IPython by Wes McKinney (ISBN-13: 978-1491957660). Available through the Purdue Library. Also available at: <https://wesmckinney.com/book/>

4. Additional reference materials are available in the Learning Sources and module sections in Brightspace.

Instructor's Response Time:

Email is the preferred method of communication. I generally respond to student questions within 24 hours during the Monday–Friday workweek. Messages sent over the weekend may receive a delayed response. *When emailing, please include the course number in the subject line to help me identify and respond to your message more efficiently.*

Online Synchronous Activity Sessions:

Each week, I will offer two live synchronous Zoom sessions. Students are encouraged to attend and participate actively with their classmates. The topic of each session will be posted in Brightspace under the Synchronous Activity section for each unit. An additional session led by the teaching assistant will also be offered; the day and time will be announced soon.

Assignments and grading schema:

Components	Percentage
Cases 3 individual cases, 10% each	30%
Discussion forum 3 individual discussion participations, 5% each	15%
Final group project One mid-term submission: 20% and one final: 25%	45%
Final reflection	5%
Individual professional portfolio on GitHub	2.5%
Set up accounts for GenAI interfaces/Configure your GitHub account	2.5%
	100%

Cases: There will be three cases in the form of written, individual homework assignments. The due dates posted on the course website are in Eastern Standard Time (EST). Late homework will **NOT** be accepted without prior permission. Some collaboration on homework is acceptable, but each student must do his/her own write-up of the solution to show their full understanding. Direct copying of another student's solution will result in grades of zero for **both** students.

Discussion forum: Students will participate in three individual discussion forum activities throughout the course. These discussions are designed to promote reflection, exchange of ideas, and application of course concepts to business analytics and AI contexts. Students are

expected to contribute thoughtful, substantive posts that demonstrate engagement with the topic and, when applicable, respond meaningfully to classmates.

Final group project: Teams will work with a real-world dataset to develop an analytics solution that integrates descriptive analysis and predictive modeling, supported by AI and LLM tools. Students will explore and visualize the data to frame a business problem, then build and evaluate predictive models for a relevant outcome. Emphasis is placed on sound analytical reasoning, model performance, and the translation of results into clear, actionable business insights. Detailed instructions and submission guidelines will be provided in Brightspace.

Final reflection: Students will submit a personal reflection discussing how the knowledge and skills acquired in the course enhance their current or future academic and professional profile. The reflection should identify three concrete applications of course concepts in the student's current role or in a future role they plan to pursue, and outline a clear plan for continuous learning in analytics and AI as a core professional competency.

Professional portfolio on GitHub: As part of the course, students will develop a professional GitHub portfolio showcasing selected projects and analytics artifacts created during the semester. The portfolio should demonstrate technical competence, clear documentation, and effective communication of analytical results, thereby strengthening the student's professional profile and expanding career opportunities in analytics and AI-related roles.

Set up accounts for GenAI interfaces / Configure your GitHub account: Students will complete an initial setup assignment to create and configure the digital tools required for the course, including selected GenAI interfaces and a GitHub account. This component ensures that all students have access to the platforms needed for assignments, project development, and portfolio creation throughout the semester. Further instructions are provided in the assignment description.

Faculty Response Times

Response Type	Timeframe
Assignment grades	Within 7 days of the due date
Assignment feedback	Within 7 days of the due date
Discussion feedback	Within 5 days of discussion closing

Grading Policy:

The target grade distribution for all core courses is as follows: 35-40% A/A-, 50-55% B+/B, 5-10% B-, 0-5% C+ or below resulting in approximately an average Grade Point Average (GPA) of **3.35** for

each core course where the GPA is calculated as A = 4, A- = 3.70, B+ = 3.30, B = 3.00, B- = 2.70, C+ = 2.30, C = 2.00, C- = 1.70, D = 1.00 and F = 0.00.

Academic Dishonesty

Purdue prohibits "dishonesty in connection with any University activity. Cheating, plagiarism, or knowingly furnishing false information to the University are examples of dishonesty." [*Part 5, Section III-B-2-a, University Regulations*] Furthermore, the University Senate has stipulated that "the commitment of acts of cheating, lying, and deceit in any of their diverse forms (such as the use of substitutes for taking examinations, the use of illegal cribs, plagiarism, and copying during examinations) is dishonest and must not be tolerated. Moreover, knowingly to aid and abet, directly or indirectly, other parties in committing dishonest acts is in itself dishonest." [*University Senate Document 72-18, December 15, 1972*]

Gen AI Use Policy

You are encouraged to use AI tools that enhance your individual or group learning. Developing the ability to work effectively with AI is a valuable and emerging skill, and I am available to provide guidance during class sessions, office hours, or by appointment. Please follow these guidelines:

- ✓ Low effort prompts typically yield poor-quality results (as we will learn in the course)
- ✓ Use AI to support your learning by requesting explanations, examples, or clarification on concepts and methods rather than direct solutions.
- ✓ Do not accept AI outputs uncritically. You are responsible for verifying accuracy and addressing any errors or omissions.
- ✓ Always acknowledge AI use. Include a brief note at the end of any assignment explaining how AI supported your work and properly cite any AI-generated sources.
- ✓ Improper use of AI tools may be treated as a violation of academic integrity policies.

Disability Statement

Students with disabilities must be registered with the Disability Resource Center in the Office of the Dean of Students before classroom accommodations can be provided. If you are eligible for academic accommodation because you have a documented disability that will impact your work in this class, please schedule an appointment with me as soon as possible to discuss your needs.

Nondiscrimination

Purdue University is committed to maintaining a community that recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages everyone to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and

nurture diversity. Purdue University prohibits discrimination against any member of the University community based on race, religion, color, sex, age, national origin or ancestry, genetic information, marital status, parental status, sexual orientation, gender identity and expression, disability, or status as a veteran. Any student who believes they have been discriminated against may visit www.purdue.edu/report-hate to submit a complaint to the Office of Institutional Equity. Information may be reported anonymously.

Mental Health and Wellness

Purdue University is committed to advancing the mental health and well-being of its students.

Counseling and Psychological Services (CAPS)

If you or someone you know is feeling overwhelmed, depressed, or in need of mental health support, contact CAPS at 765-494-6995 (available during and after hours, on weekends and holidays).

Additional Support

- ✓ Office of the Dean of Students: Call 765-494-1747. Hours: Monday through Friday, 8 AM to 5 PM.
- ✓ Wellness Coaching: Sign up for free one-on-one virtual or in-person sessions with a Purdue Wellness Coach at RecWell. Signing ups is free and can be done on BoilerConnect.

If you are in crisis, please call or text 988 for the Suicide and Crisis Lifeline, available 24/7.

Emergency Preparation.

In the event of a major campus emergency, course requirements, deadlines, and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted in Brightspace and communicated via email. You are expected to read your @purdue.edu email on a frequent basis.

Basic Needs Support

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students for support.

- ✓ Website: www.purdue.edu/studentsuccess/basicneeds
- ✓ Email: basicneeds@purdue.edu

✓ Phone: 765-494-1747

Netiquette

Your instructor and fellow students wish to foster a safe online learning environment. All opinions and experiences, no matter how different or controversial they may be perceived, must be respected in the tolerant spirit of academic discourse. Guidelines include:

- ✓ Do not dominate any discussion. Allow other students to join in.
- ✓ Do not use offensive language. Present ideas appropriately.
- ✓ Be cautious in using Internet language. Capitalizing all letters suggests shouting.
- ✓ Keep an open mind and be willing to express even your minority opinion.
- ✓ Think and edit before you push the "Send" button.

Course Schedule

The instructor reserves the right to make changes to the schedule or syllabus with notice to students. Any changes will be announced in class and/or posted in Brightspace.

MODULE	WEEK	TOPIC(S)	READINGS /VIDEOS	ASSIGNMENTS
Module1. The AI ecosystem	W1	1.1 Why AI Matters for Business Analytics 1.2 AI vs. ML vs. DL vs. GenAI 1.3 Large Language Models (LLMs) 1.4 How LLMs Work: an overview.	Set of videos and readings: W1	- Set up accounts for GenAI interfaces. - Configure your GitHub account. Due Jul 5, 11:59 pm (EST)
Module 2. Prompt Engineering and Algorithmic Thinking	W2	2.1 Communicating with LLMs 2.2 Prompt Engineering Fundamentals 2.3 Algorithmic Thinking 2.4 Prompting for Analytics	Set of videos and readings: W2	Discussion 1: Prompt Engineering. Due Jul 12, 11:59 pm (EST)
Module 3. Describing,	W3	3.1 Data exploration and wrangling		- Case 1 NovaRetail. Due Jul 19, 11:59 pm (EST)

MODULE	WEEK	TOPIC(S)	READINGS /VIDEOS	ASSIGNMENTS
exploring, and visualizing data		3.2 Data visualization 3.3 Dashboards and descriptive analytics	Set of videos and readings: W3	'- Final Project mid-term submission Due Jul 27, 11:59 pm (EST)
Module 4. Supervised Learning: Predictive modeling.	W4	4.1 Predicting numerical/categorical outcomes 4.1.1 Prediction, performance, and error 4.1.2 Models 4.1.3 Fine-tuning and model selection	Set of videos and readings: W4	Case 2 Airbnb. Due Jul 27, 11:59 pm (EST)
Module 5. Unsupervised Learning: Clustering	W5	5.1 Clustering: goals and business benefits 5.2 Distance measures 5.3 Profiling and description	Set of videos and readings: W5	- Discussion 2: Predictive Analytics. Due Aug 2, 11:59 pm (EST) - Case 3 IBM Clustering. Due Aug 4, 11:59 pm (EST)
Module 6. Agentic AI	W6	6.1 AI Agents and the Agentic Perspective 6.2 Agent Construction and RAG. 6.3 Integrating Agents into Data-Driven Solutions	Set of videos and readings: W6	- Discussion 3: Responsible AI. - GitHub Portfolio ready and uploaded Both due Aug 9, 11:59 pm (EST)
Module 7. Ethics and responsible AI.	W7	7.1 Bias, Fairness, and Risk in AI 7.2 Transparency and Explainability 7.3 Assessing Fairness in Data Solutions	Set of videos and readings: W7	- Final (individual) reflection. - Final Project submission. Both due Aug 15, 11:59 pm (EST)